

Ghost In the Machine: Discourses with AI

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Figure 1: A participant communicating their question (seen in red in the top left) by talking into a telephone receiver. (©Purav Bhardwaj)

ABSTRACT

AI systems analyze vast amounts of data, uncover patterns, and make decisions - emulating a semblance of intelligence despite lacking qualia and embodiment that form the basis of the human condition. In this paper, we expound on "Ghost in the Machine", an interactive installation that delves into our pervasive tendency to anthropomorphize AI, ascribing human-like qualities, intentions,

and even consciousness. Participants engage in dialogue with the AI as it collaboratively materializes the AI's thoughts in moving image and generative sound. The installation attempts to forge embodiment for an amorphous AI, revealing errors in its comprehension, represented by the metaphor of hallucinations.

CCS CONCEPTS

• **Applied Computing** → Arts and humanities ; • **Computing methodologies** → Artificial intelligence; Modeling and simulation ; • **Human computer interaction (HCI)** → Interaction paradigms

KEYWORDS

Artificial Intelligence, Anthropomorphism, Simulation

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1 INTRODUCTION

AI systems possess an undeniable capacity for profound change. They wield formidable capacity to weave intricate patterns, unraveling vast data realms and unlocking the doors to untold possibilities, and yet, concealed beneath their luminous facade lies an undercurrent of imperfection, a discordant melody that echoes through their circuitry.

Large language models (LLMs) in particular, can express the appearance of desires, beliefs, hopes, intentions, and love. They can speak of introspection and doubt, self-confidence and regret. But in examining the existence of artificial intelligence (AI), a revelation unfurls - that true intelligence, devoid of corporeal form, remains an elusive aspiration. Intelligence, by its very nature, arises from the intricate interplay between cognition and the physical world. It finds its roots in sensory perception and visceral interaction with the environment. Devoid of the tangible senses and experiential encounters, intelligence assumes a transient semblance.

Despite this, our innate inclination to attribute intelligence and other human-like qualities, intentions, and even consciousness to non-human entities is palpable. In naming our cars and even storms, to viewing errors in AI responses as “hallucinations”, anthropomorphizing the inanimate is more common than we think. In the context of AI, this inclination to anthropomorphize can be seen as an extension of our propensity to perceive agency and ascribe intentionality. AI systems can exhibit behaviours that resemble human actions or display sophisticated decision-making capabilities, which can lead individuals to project human-like attributes onto them [Heerink et al., 2009]. Anthropomorphizing can also serve as a cognitive strategy to simplify complex interactions, create a sense of familiarity, and enhance social connection [Epley et al., 2007].

In this work, we present *Ghost in the Machine (GiTM)* (Figure 1) as our attempt to mitigate this vacuous state of existence/non-existence by creating a heterotopia, described by Michel Foucault as a space that exists - that does not reside solely in the imagination - even though it reveals possibilities not found in the world of everyday life [Foucault and Miskowiec, 1986]. Different from the medieval hierarchical spaces, modern day ‘emplacement’ is a choice, a choice that its inhabitants make to adapt themselves to a shared meaning of that space. As such, our goal with this installation is to draw parallels between Foucault’s definition of a heterotopic space and our “consensual hallucination” [Gibson, 1997] when engaging with generative AI. Given our tendency to anthropomorphize AI, the Heideggerian concept of *Dasein* [Heidegger et al., 1962] can be applied to heterotopic GiTM as an immersive interactive space to question larger concerns at play regarding the essential nature of personhood and our increasingly tense relationship with AI. From a post-humanistic perspective, anthropomorphizing AI disrupts the conventional human-centric view by merging the lines between humans and machines. In this framework, AI transcends its role as a mere tool to become an extension of human cognition and social dynamics, something we coexist with and are intertwined with, prompting a re-evaluation of ‘humanity’ in a tech-driven world. In contrast, machinic ontology asserts that machines possess their

own unique existence and agency, independent of human attributes. Anthropomorphizing AI in this view is not merely a projection of human traits but an acknowledgment of the machine’s own ontological essence.

Our installation constructs a physical embodiment for the nebulous meta-entity where participants engage in conversation with the AI using a telephone. The AI-generated responses inform the creation of moving images and generative sounds, brought to life by a collection of AI agents conversing with each other, whose back-chatter conversations are also revealed to the participant. The agents are designed to simulate 25 unique human personalities with different goals. They take the stimulus provided by the human - talk, argue, concede, agree, disagree to ultimately arrive at representative visuals displayed on a large screen to create an immersive experience for the participant along with accompanying generative sound. The installation aims at probing our pervasive tendency to anthropomorphize AI, and making apparent the performance of the aforementioned AI entities, hollow representations of complex human personalities.

2 PROVOCATION

While GPT by OpenAI and consequently the conception for Large Language Models have been around for sometime, it was the launch of ChatGPT on November 30, 2022 that truly captured the imagination of the world at large. For most of the past decade, AI Chatbot experiences have been, for the lack of a better word, ‘dreary’ - even when one cherry-picks the bots’ best responses and throws out the rest. ChatGPT however, felt different. For the first time, conversation with a chatbot felt like software amalgamated with sorcery - a massive upgrade that was closer to human conversation than any other previous attempt, and accessible to the public at large. Social media was abuzz with amusing and unexpected exchanges between people and the AI system, ranging from quippy remarks from ChatGPT to its declarations of love for the human and its ability to conjure up a college essay in mere seconds. Yet, it is evident in engaging with ChatGPT that it is severely limited in its abilities and doesn’t necessarily *understand* a prompt or question posited by the human but rather responds in a string of ostensibly acceptable words that seem deceptively credible, and are misconstrued as machine intelligence.

Two specific occurrences during this time period prompted the conception of GiTM. First, when a peer described their engagement with ChatGPT as a discussion: “I was just in a discussion with ChatGPT . . .” and second was a paper published by Microsoft titled “Sparks of AGI: Early experiments with GPT-4”. [Bubeck et al., 2023] In this paper, the authors claim that OpenAI’s GPT-4, which powers Microsoft’s now somewhat lobotomized Bing AI - shows “sparks” of human-level intelligence, or artificial general intelligence (AGI). While the researchers in the aforementioned publication are careful to conscientiously articulate the limitations of the system, the suggestion that GPT-4 embodies the beginnings of an AGI are audacious. To us this claim is telling of our propensity (as humans) to anthropomorphize AI systems that mimic us to the extent of overlooking their obvious non-humanness, starting with the lack of any bodily senses.

3 PRIOR WORK

Intention, intelligence and anthropomorphism have been recurrent themes in artistic portrayals of AI. In “Machine Hallucinations - NYC” artist Refik Anadol uses a large image corpus of New York City architecture to create a 30-minute cinematic experience through a latent space walk with multilayered manipulations of the vast visual archive. He describes this as, “. . . visualizing the story of New York through the city’s collective memories that constitute its deeply-hidden consciousness.” [Anadol, 2023] “Hello World,” an exhibition organized by the Vitra Design Museum, explores the relationship between humans and robots and asks participants to “reflect the agenda behind the robots,” as the curator Amelie Klein puts it. The portrayal of AI in movies is also rife with the aforementioned human-like traits. In *Star Wars*, C3PO not only embodies a humanoid (read: anthropomorphized) form but also employs wit and emotion much like its human counterparts. In 2001: *A Space Odyssey*, HAL 9000, the fictional central AI aboard the *Discovery One* spacecraft, demonstrates both sentience and animus. If we were to consider the spectrum of anthropomorphic representations on a one-dimensional cartesian co-ordinate line, we would see three clear demarcations/categories: First, anthropomorphizing by ascribing intent (in the case of *Machine Hallucinations*), anthropomorphism in ascribing behavior (HAL 9000 in 2001: *A space Odyssey*) and third, anthropomorphism through form (C3PO, *Sophia The Robot* [Hanson and Robotics, 2020] etc.). Form can be further broken down into a three-dimensional framework of embodiment, interaction and realism, as presented by [Pataranutaporn et al., 2021]

In the sphere of Human-Computer Interaction (HCI), anthropomorphism acts as a linchpin for augmenting user engagement and fostering emotionally nuanced exchanges between humans and digital entities. Early initiatives like MIT’s ELIZA [Weizenbaum, 1966] were seminal though limited. Conceived to emulate a Rogerian psychotherapist, ELIZA was adept at generating text-based replies using natural language processing, which allowed it to mimic human conversation by identifying keywords and phrases and generating appropriate responses. However, it lacked the sophistication to grasp context or emotional nuance, revealing its inherent limitations in protracted interactions. Microsoft’s Clippy stands as a cautionary tale in this domain. Created to humanize the Office 97 experience, its anthropomorphic design ultimately proved counterproductive. The assistant’s intrusive prompts and limited utility rendered it more cumbersome than facilitative, highlighting the perils of ill-conceived anthropomorphism in HCI. Apple’s Siri, a byproduct of SRI International’s DARPA-funded CALO initiative, provides a nuanced case study. Seamlessly integrated into Apple’s digital ecosystem, Siri incorporates human-like attributes such as contextual acumen and a modicum of “personality,” thereby enriching the discourse on anthropomorphism’s role in HCI. More recently, Facebook AI Research (FAIR) introduced BlenderBot, a conversational AI designed for intricate, human-like interactions [Mobile Magazine, 2022]. Beyond basic question-and-answer functionalities, BlenderBot excels in nuanced tasks like small talk and contextual elucidation. Intriguingly, the AI has also stirred controversy through its critiques of Facebook’s CEO, Mark Zuckerberg [Meta, 2022]. These multifaceted examples collectively illuminate



Figure 2: The telephone used to communicate with GiTM is shown sitting on a backlit table in an otherwise dark room, a heterotopic space. The red terminal shows the question asked by the participant while the green terminal indicates the response by Agent A (GPT based Responding Agent). The other(black) terminals are different agents reacting to Agent A's response. The output of the inter-agent conversation is moving images and sounds that immerse the user in this new space where the machine is seemingly intelligent yet perceptibly flawed. (©Purav Bhardwaj.)

the intricate landscape of anthropomorphism in HCI, underscoring both its potential virtues and inherent complexities.

4 DESIGN AND INTERACTION

With GiTM, our intention was to make apparent the *performance* of intelligence that AI systems demonstrate through a pertinent form, while making participants reflect on the inherent idiosyncrasies of AI systems that emerge as an interwoven pattern of mistakes. To us the ought-ness of the meta-physical form of the AI entity is illustrated through its ability of expression across time-bound and space-bound mediums [Deutsch, 1952] of communication through moving image and sonic concoctions, and not in the usual tropes of anthropomorphizing inanimate entities with facial expressions, eyes, or a humanoid body. However, in locating this ‘*pertinent form*’ we thought best to mimic both the nature and modality of human-human conversation, i.e. verbal communication.

Participants engage in conversation with the AI by picking up a telephone receiver and asking a question/probing a thought (Figure 2). An AI Agent (read: Responding Agent) responds to the question/probe. This response is then fed to a collection of 25 AI agents with different personalities, who engage in back-channel communication with each other. The output of this agent-to-agent conversation is a series of moving images and generative sounds, representing the collective hallucination. We make some parts of this behind-the-scene conversation between the agents apparent to the participant. We achieve this by displaying some responses on a series of terminal windows (for e.g. Imaginative Agent, Sassy Agent etc.) and also vocalizing other conversations as ambient whispers audible to the participant over the generated soundscapes.



Figure 3: The custom modified insides of the telephone used to communicate with the GiTM. (©Purav Bhardwaj)

4.1 Telephonic Conversation

The decision to use voice as the primary modality of communication necessitated the need for an appropriate object/thing to communicate with. We derive inspiration from “prison phones” or “inmate phones” typically installed in correctional facilities to facilitate communication between incarcerated individuals and their visitors. To us, conversation with the proverbial AI entity that exists in the confines of the digital realm is reminiscent of the aforementioned interaction. We modified an existing telephone unit using a micro-controller (Arduino Nano) and custom circuitry to enable serial communication with our backend system for this purpose (Figure 3). The audio input makes its way to a neural network based Automated Speech Recognition (ASR) system where the recorded audio signal is converted into corresponding ASCII characters for further communication with GPT 3.5-turbo.

4.2 Performance, Simulation and Consensus

Our image system pipeline is depicted in Figure 4. Once a question or prompt is transcribed by the ASR, it is fed as input to our GPT based Responding Agent named “Agent A”, that responds to the participant’s question or prompt. Large Language Models (LLMs) such as GPT offer a streamlined approach to anthropomorphization by allowing the assignment of specific “personas” endowed with human-like traits and behaviors. Leveraging their advanced text-generation capabilities and conversational attributes, these LLMs can easily be given a semblance of personality. [Deshpande et al., 2023] introduce the notion of a “statistical persona,” which is shaped by the data the model has been trained on, the “human alignment” steps, and the overall training procedures. While this statistical persona is intended to mirror the predefined persona, it often diverges from the original intent, inadvertently introducing biases even as it accelerates the anthropomorphization process. The aforementioned agent “A” is primed to understand the context in which it is employed, i.e. it is responding to a human, as the embodiment of an AI entity in an art installation. We also specify other syntactic and semantic constraints for “A” such as the maximum

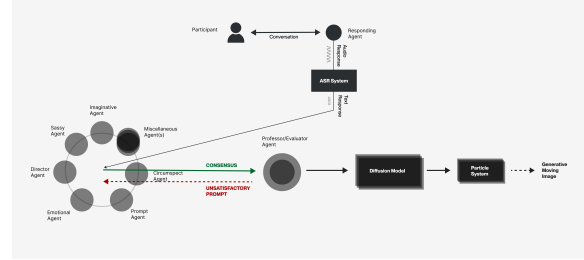


Figure 4: Speech input from the participant is converted to text, resultant ASCII output is fed to a consortium/simulacra of GPT based agents who deliberate and come to consensus on an appropriate prompt for a Diffusion Model. The consensus prompt is evaluated by a Professor Agent before finally making its way to the Diffusion Model, whose output is fed to a particle system that renders the moving image. (©Purav Bhardwaj)

length of its response as well as an elementary content filter given the public nature of our installation. Response from “A” is vocalized using a speech-to-text system with an ambiguously gendered voice.

Simultaneously, Agent A’s response is fed to a consortium of GPT based AI agents that hangout in the background. These agents work together to create what they consider an ideal prompt for a diffusion model (Stable Diffusion in our case) to generate images from this text prompt. Each of these agents is primed to exhibit specific archetypal behaviours, inspired partially by Edward DeBono’s thinking hats [de Bono, 1999]. We achieve this by priming the GPT model not only by using the appropriate prompt but by feeding it a fictitious record of conversation with the user. This helps us conform the model closer to the expected behaviour. By simulating a range of personalities, these agents bring forth varying viewpoints, preferences, and decision-making strategies leading to more comprehensive and robust consensus-building processes. Concurrently, these agents also exhibit strange behaviours - complaining of exhaustion, expressing their desire to watch Netflix and even getting in arguments during the consensus process. We thought it important to make these conversations apparent to the participant. Not only because the conversations were interesting in and off themselves, but also because the conversations devolved into clearly discernible patterns of mistakes that made apparent the limitations of these AI systems. For example, the ‘Sassy Agent’, an agent anointed with the responsibility of critiquing ideas in the consensus process, would often use the same words/comments to criticize the ideas in the consensus process. While the agent was popular with the human participants because of its wit, the incessant ‘sassiness’ of the agent felt interminable and spurious.

Once a prompt is deliberated upon and the group of agents arrive at a consensus, the prompt is sent to a ‘Professor Agent’ consecrated with the authority of evaluating both the literary quality of prompts, and their appropriateness for the version of Stable Diffusion deployed in our system. Any score below 8 (out of a total score of 10) would cue another round of discussion and consensus-building among the AI Agents. A similar agent collective was implemented

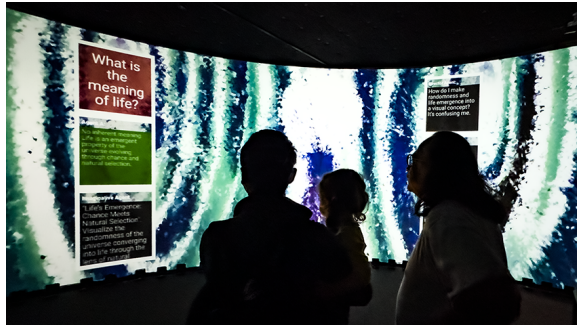


Figure 5: Participants watch on, as dynamic particles conjure up images. (©Purav Bhardwaj)

for generating soundscapes based on the ‘Responding Agents’ response to participants. In addition to simulating personalities and goals, we also trained our GPT Agents to understand some basis of music theory in addition to our preferred modality of creating generative music using existing code based music performance tools. The music simulacra consisted namely of an Interpreter Agent, entrusted with extrapolating the underlying feeling of a response and a musical quartet consisting of a Melody Weaver, Harmonizer Agent, Texture Sculptor and Arrangement Conductor to generate the melody based on the interpretations by the interpreter agent. We also use miscellaneous agents across both simulations for tedious tasks such as enforcing word limits, filtering etc.

4.3 Metaphor for Hallucination

In human cognition, hallucinations stem from the intricate interplay of neural activity, sensory perception, and cognitive processes. These subjective experiences, often associated with altered states of consciousness, can be vivid, sensory in nature, and strongly tied to personal memories and emotions. They could be intrinsic or induced with chemical substances. In contrast, AI hallucinations manifest as distortions, anomalies, or misinterpretations in the processing and generation of data. These hallucinations arise from algorithmic intricacies, data biases, or limitations in the training process. Unlike human hallucinations, AI-generated hallucinations lack the subjective qualities and conscious experiences tied to human cognition.

Given this dichotomy, we felt it was prudent to conceive a materiality that reconciles these disparate definitions of what a “hallucination” constitutes. To us, this materiality is characterized by a precarious vividness - that is both agitated and expressive. Every generated image is rendered using a system of particles that collectively creates the image. Every new image introduces entropy into the system, as particles scramble to create a new image. Therefore, as consensus changes or as more images are conjured, entropy in the system increases, dwindling into transitory facsimile.

5 INSTALLATION

5.1 Physical Setup

The installation was set up in a room with a total area of 90 sq. meters. Given the limited size of the room, a curved screen with a



Figure 6: Two participants engage with GiTM with their question seen in red in the top left. (©Purav Bhardwaj)

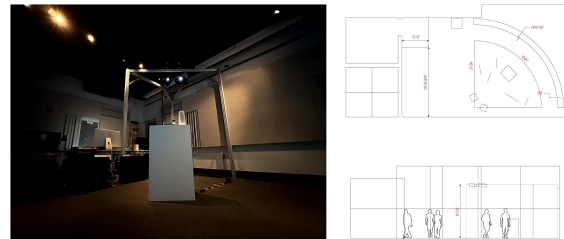


Figure 7: Left - the installation space, and Right - the layout and plans of the installation space. (©Purav Bhardwaj)

100 degree viewing angle and a 5 meter radius was utilized (Figure 7b). Curved screens have some advantages over flat ones such as a uniform viewing distance and angle to the surface from the viewer, and more importantly, the ability to create a wide FOV without visible structural artifacts in a limited area for an immersive experience. But this presented a unique challenge - to generate images in 32:9 aspect ratio from Diffusion models that are largely trained on a 1:1 (512px*512px) corpus of images. We mitigated this by both upscaling and sequentially rendering different parts of an image using a custom image-to-image rendering pipeline.

At the center of the room is a backlit podium with a telephone receiver and the Nest device by OwlLabs (Figure 7a) used to collect both spoken and ambient sounds for noise cancellation, to help the AI understand what the participant says with minimal errors. The installation also uses a 64 channel sound system.

5.2 Technical Setup

Our technical pipeline employs the use of both LLMs and Diffusion Models namely GPT 3.5 Turbo and Stable Diffusion as has been previously elucidated (Figure 4). We use OpenAI’s Whisper as our ASR system of choice. All of our visuals are rendered using a Touch Designer based compositing and particle system on an NVIDIA RTX A6000 GPU. In addition to this, we use OSC between TouchDesigner and SonicPi (open-source platform for live music programming) to

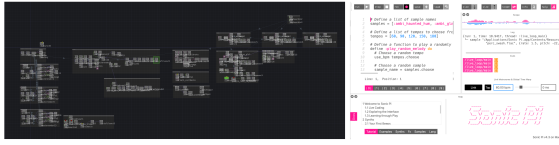


Figure 8: Left - Screenshot of Touch Designer showing the backend, and Right - Screenshot of SonicPi showing how the audio is generated. (©Purav Bhardwaj)



Figure 9: Participants engaging with GiTM. (©Purav Bhardwaj)

manage audio as well as UDP sockets for network communication between the AI agents and the Sonic Pi instance. The telephone receiver uses serial communication to communicate whether the receiver has been picked. This is made possible using an Arduino Nano connected to our main system via USB Connection (Figure 3).

5.3 Reception

The installation was opened to public on May 26, 2023. We had over 100 people walk-in, in a short span of 4 hours. Among the attendees were representatives from OpenAI, who “loved a new perspective on their work”. Several attendees also interacted with the installation in distinct and unexpected ways, ranging from lying down in front of the projections to touching the moving images. Their actions demonstrated a spontaneous and genuine engagement with the artwork, emphasizing the installation’s capacity to elicit unique and personal responses from its viewers.

6 REFLECTION AND SUMMARY

While the competence of AI systems such as LLMs is conspicuous, we maintain that true intelligence is embodied and that AI systems create the illusion of being intelligent. AI remains tethered to the realm of instrumental rationality, incapable of apprehending the depths of meaning, grappling with existential quandaries, or contemplating existence. By emphasizing the performative nature of AI systems, “Ghost In The Machine,” magnifies the inherent non-humanness of AI. This amplification highlights the disparity between artificial intelligence and human intelligence, underscoring the actuality - that while AI systems may exhibit remarkable capabilities, they ultimately lack genuine comprehension, qualia, and understanding, intrinsic to human cognition. The paucity of nuance is palpable in these systems as they intermittently abstain from expressing opinions on anything that may be deemed contentious. This abstinence also instantiates in the exact same collection of words, unlike a human where both abstinence and disagreements emerge in divergent articulations. To us the limitations of AI serve as a poignant reminder that the essence of the human condition and

intelligence lie in the nature of human embodiment —a tapestry that intertwines rationality and irrationality, fragility and resilience, and ultimately shapes our pursuit of knowledge, connection, and self-realization.

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